

Ethanol Blending in Gasoline - Honduras

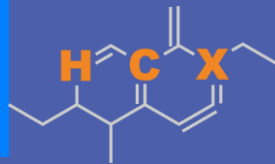
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Ethanol Blending in Latin America

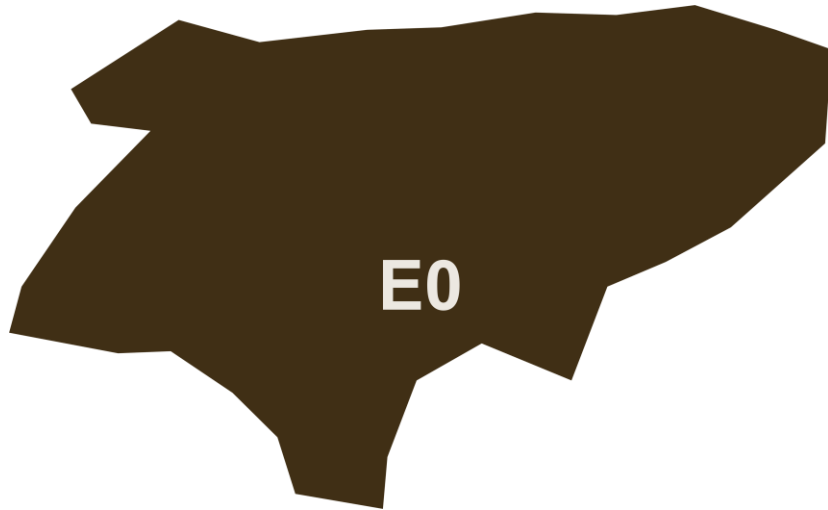
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Honduras



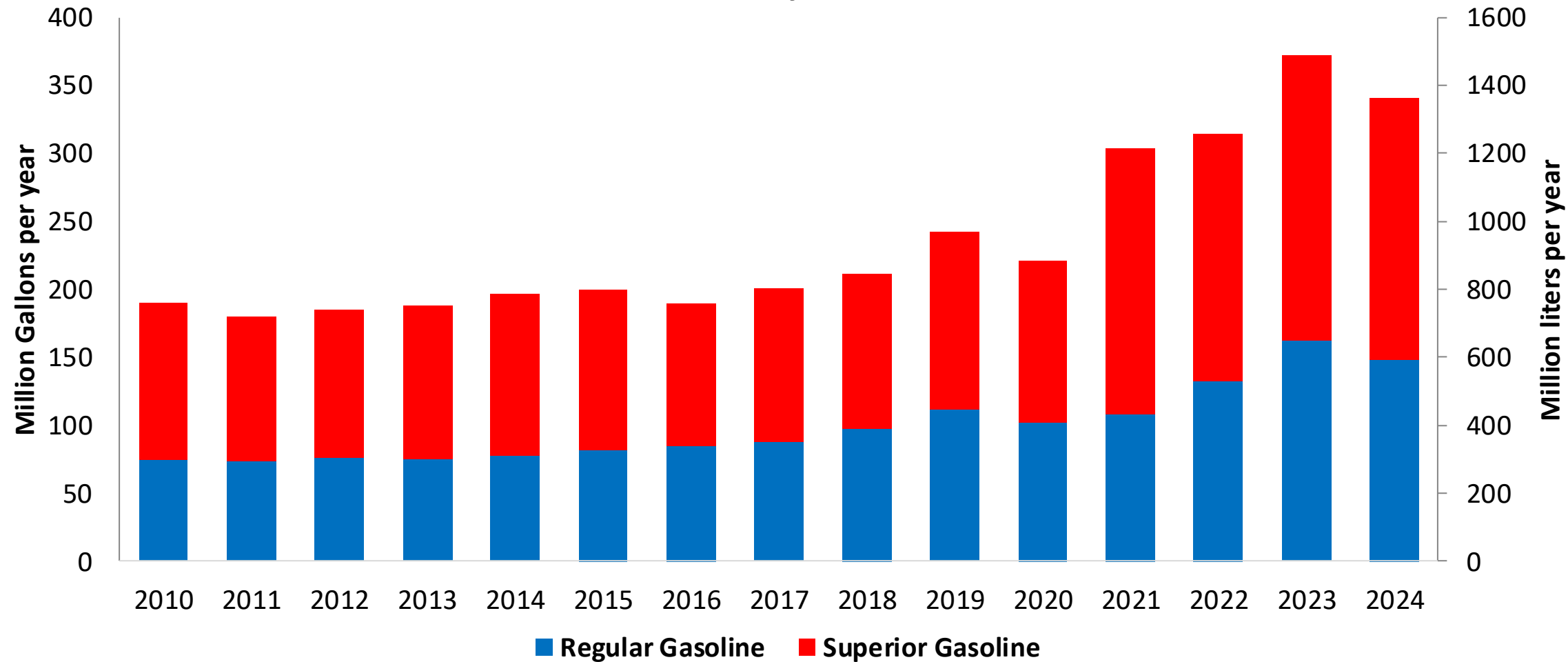
In 2024, gasoline demand was 374 million gallons (1,290 million liters). Regular gasoline RON 91 represented 46% while Superior gasoline RON 95 reached 54% of national demand. Honduras does not have gasoline production capacity and supplies its market with imports mainly from the United States.

Honduras has no national mandate for ethanol blends. A new biofuel's law is under discussion.

Source: CEPAL, Secretaría de Energía Honduras

Gasoline Demand in Honduras

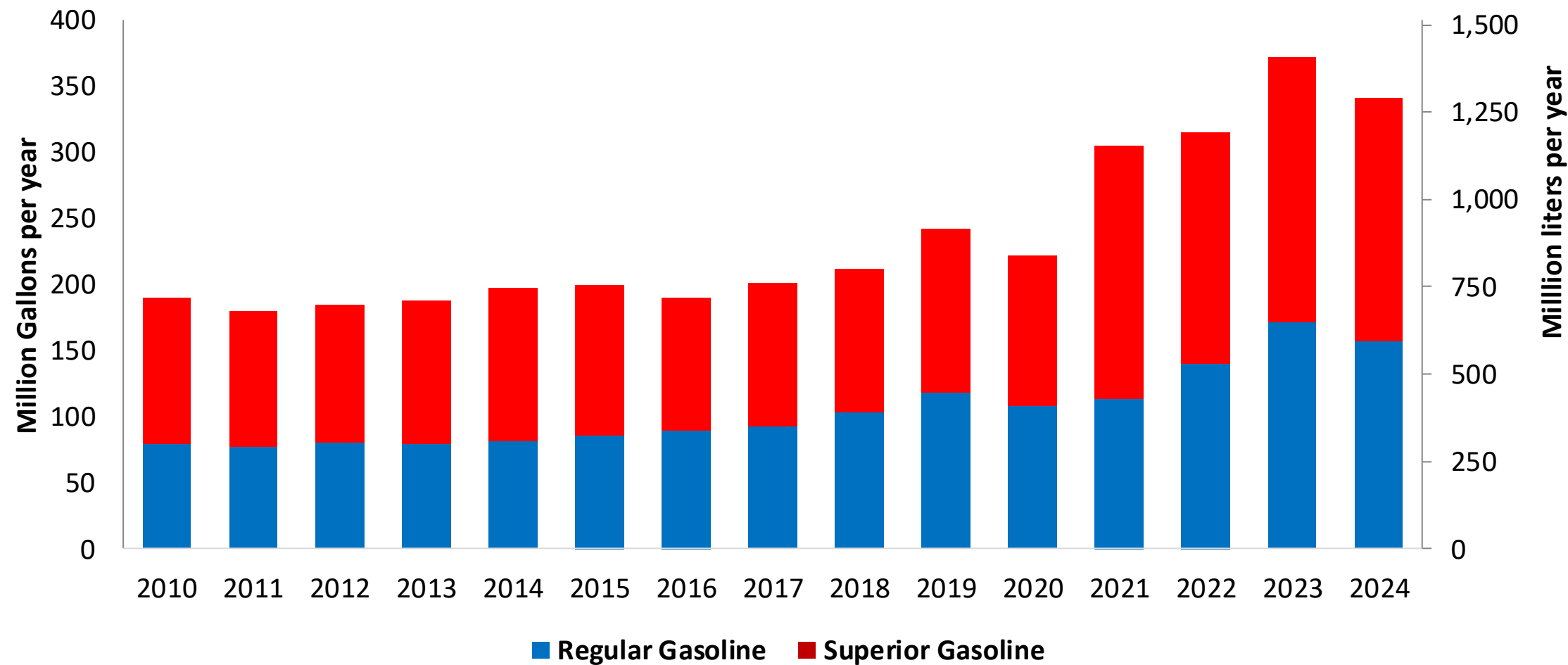
Gasoline Demand by Grade in Honduras



Source: CEPAL, Secretaría de Energía Honduras

Gasoline Imports in Honduras

Gasoline Imports by Grade to Honduras



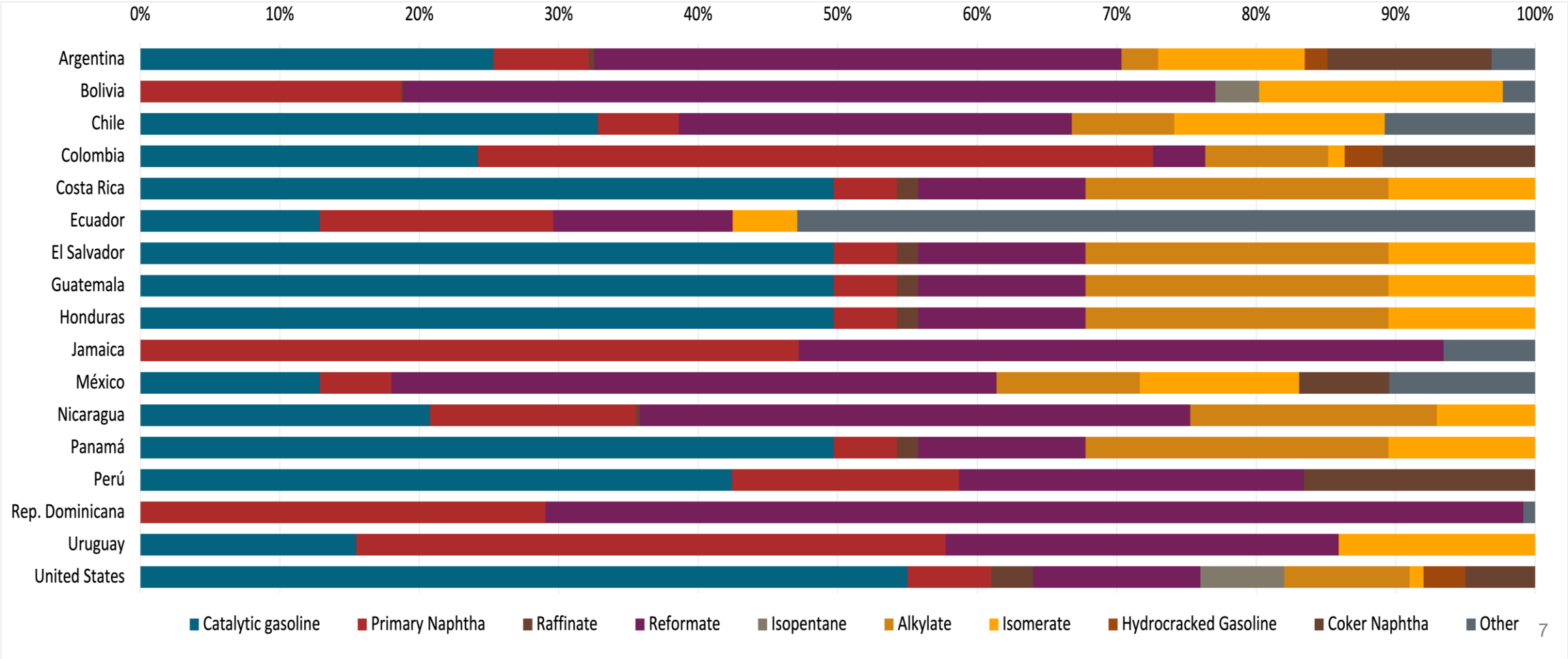
Source: CEPAL, Secretaría de Energía Honduras

Gasoline Quality in Honduras

Name	RTCA 75.01.08:19 p.m.	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	2,5% v/v	2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0,25% v/v	0,25% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



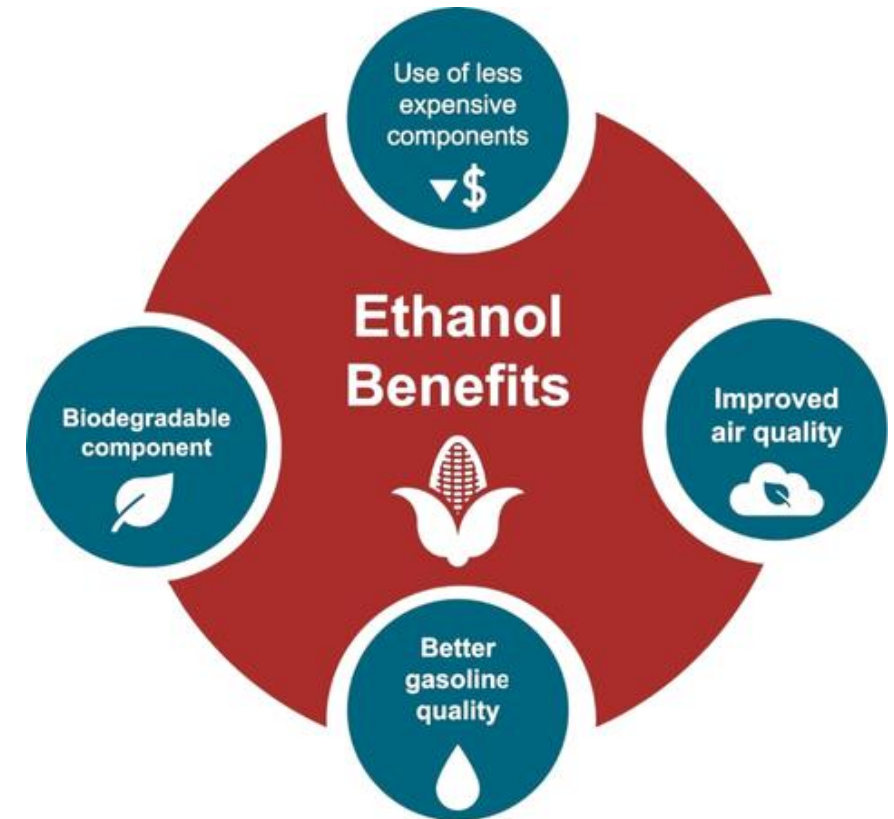
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

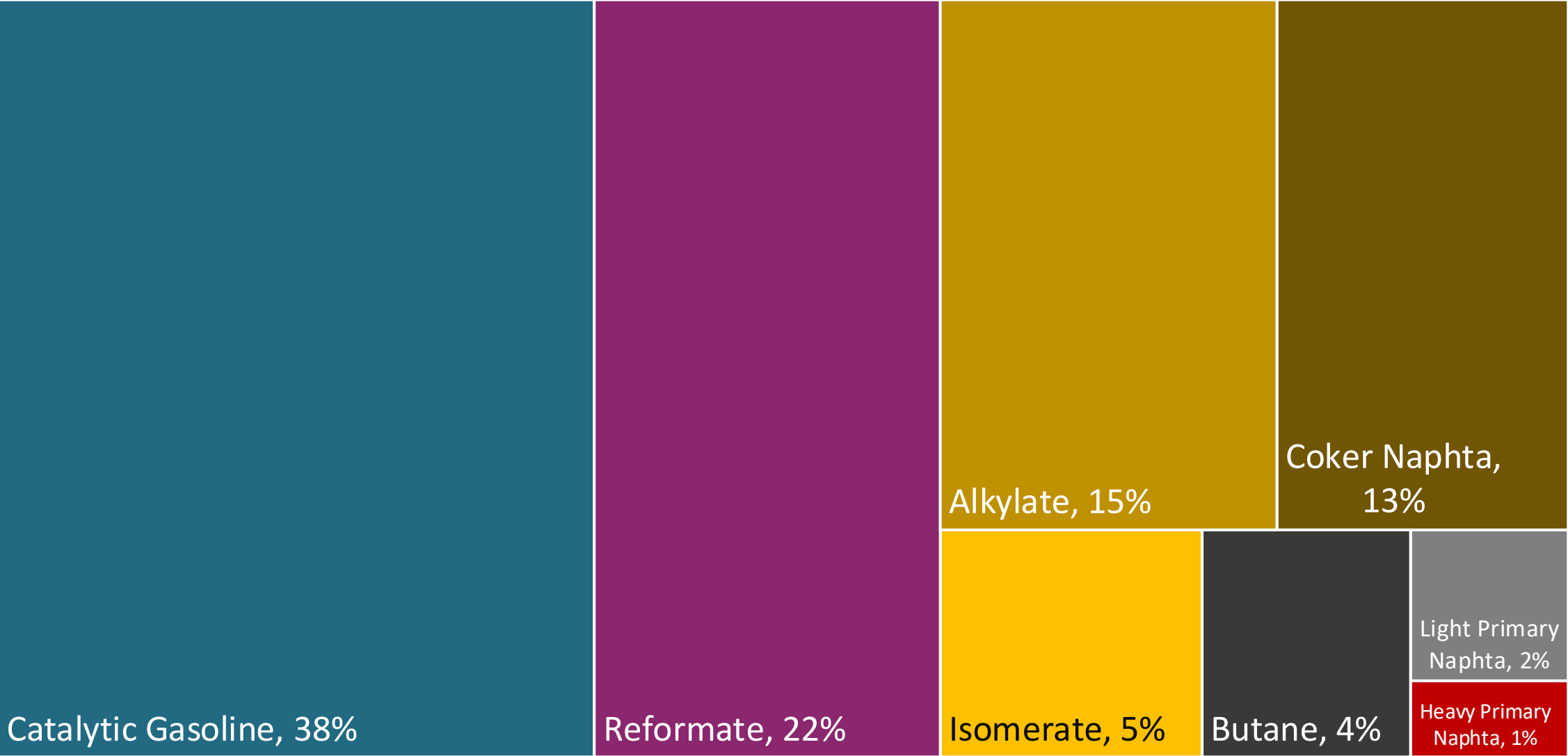
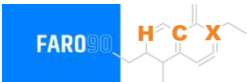
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

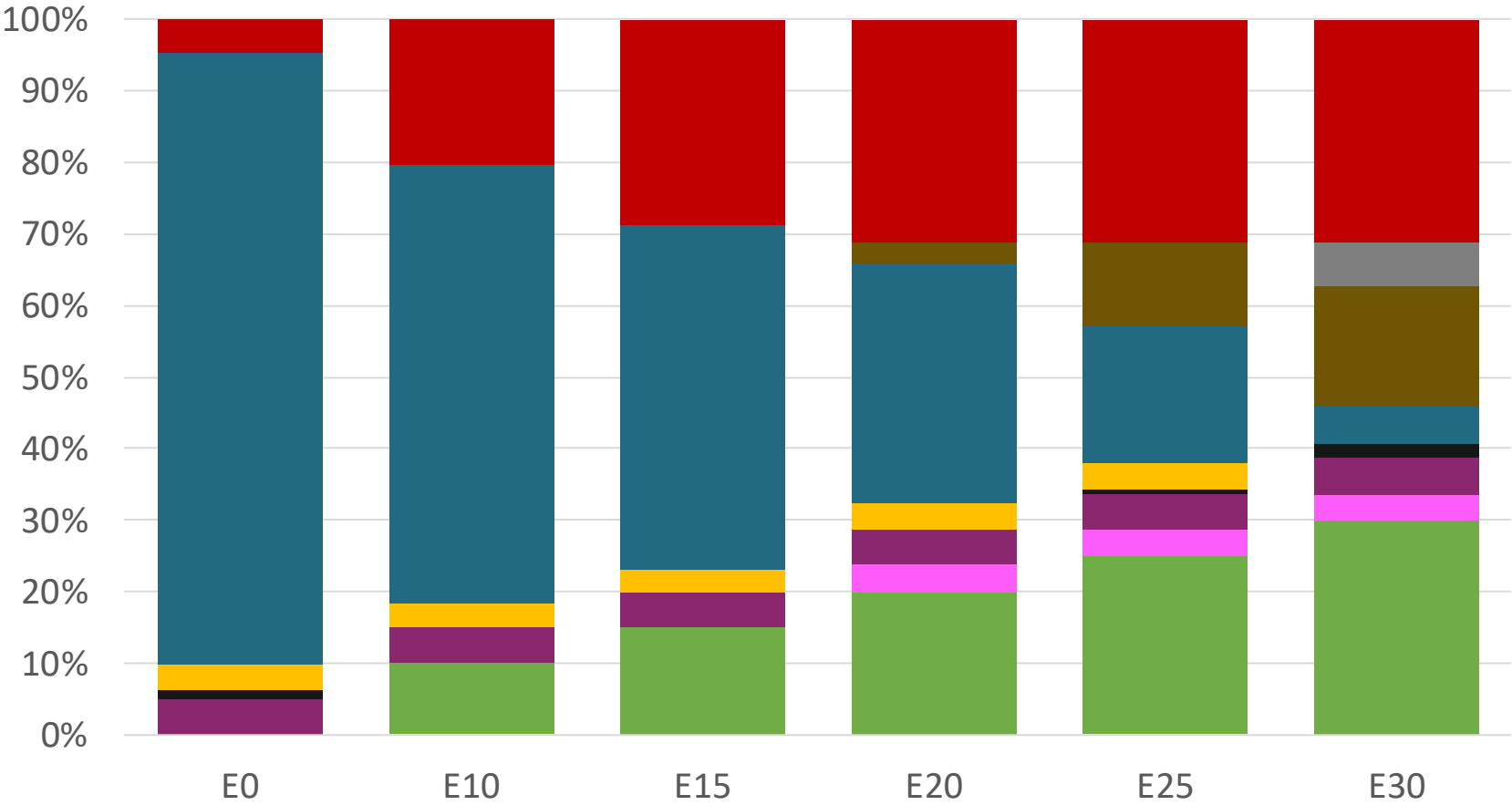
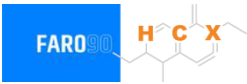


Blending Components - Honduras



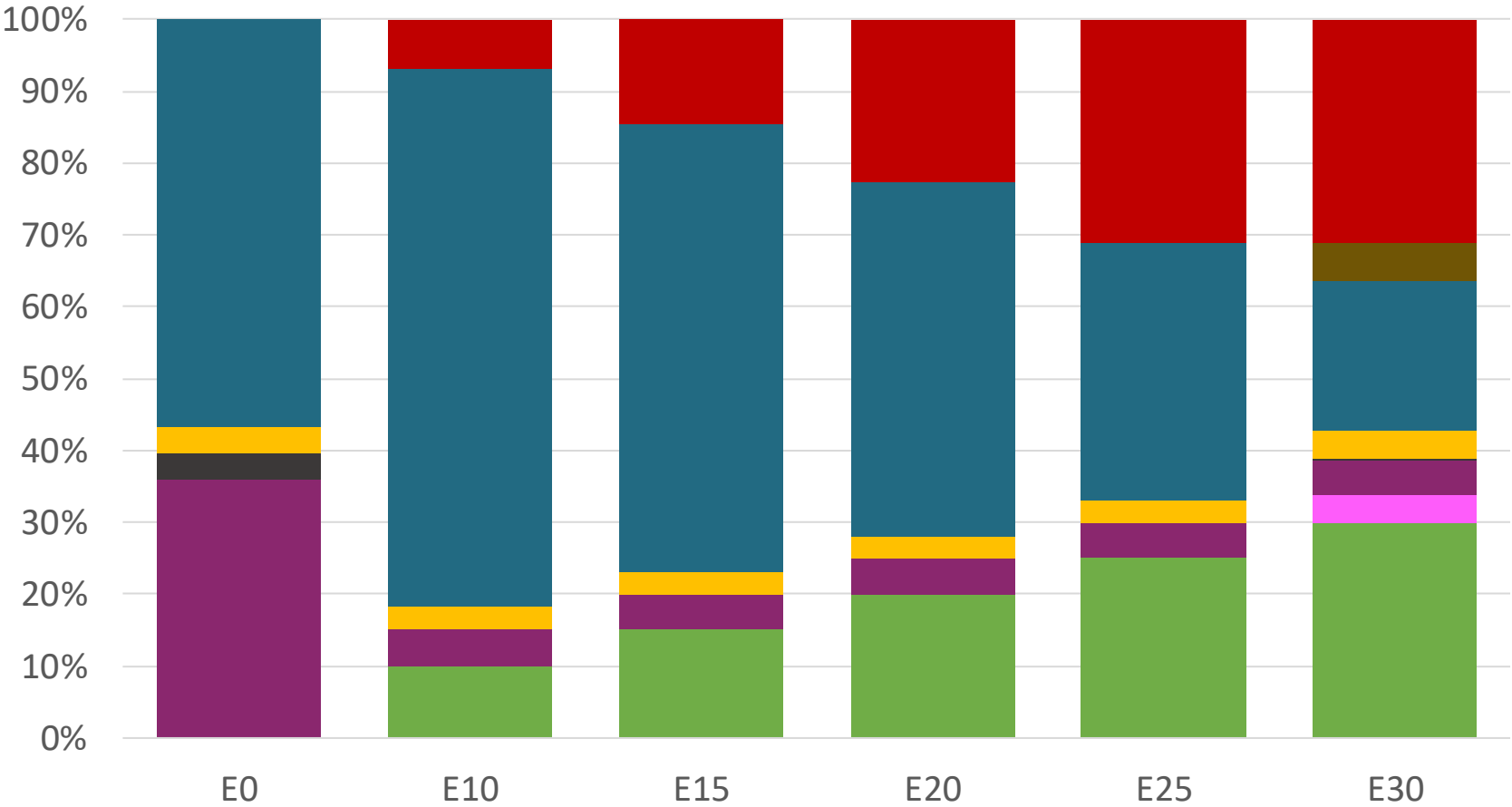
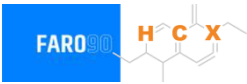
Source: Faro90

Honduras – Regular – Constant Octane



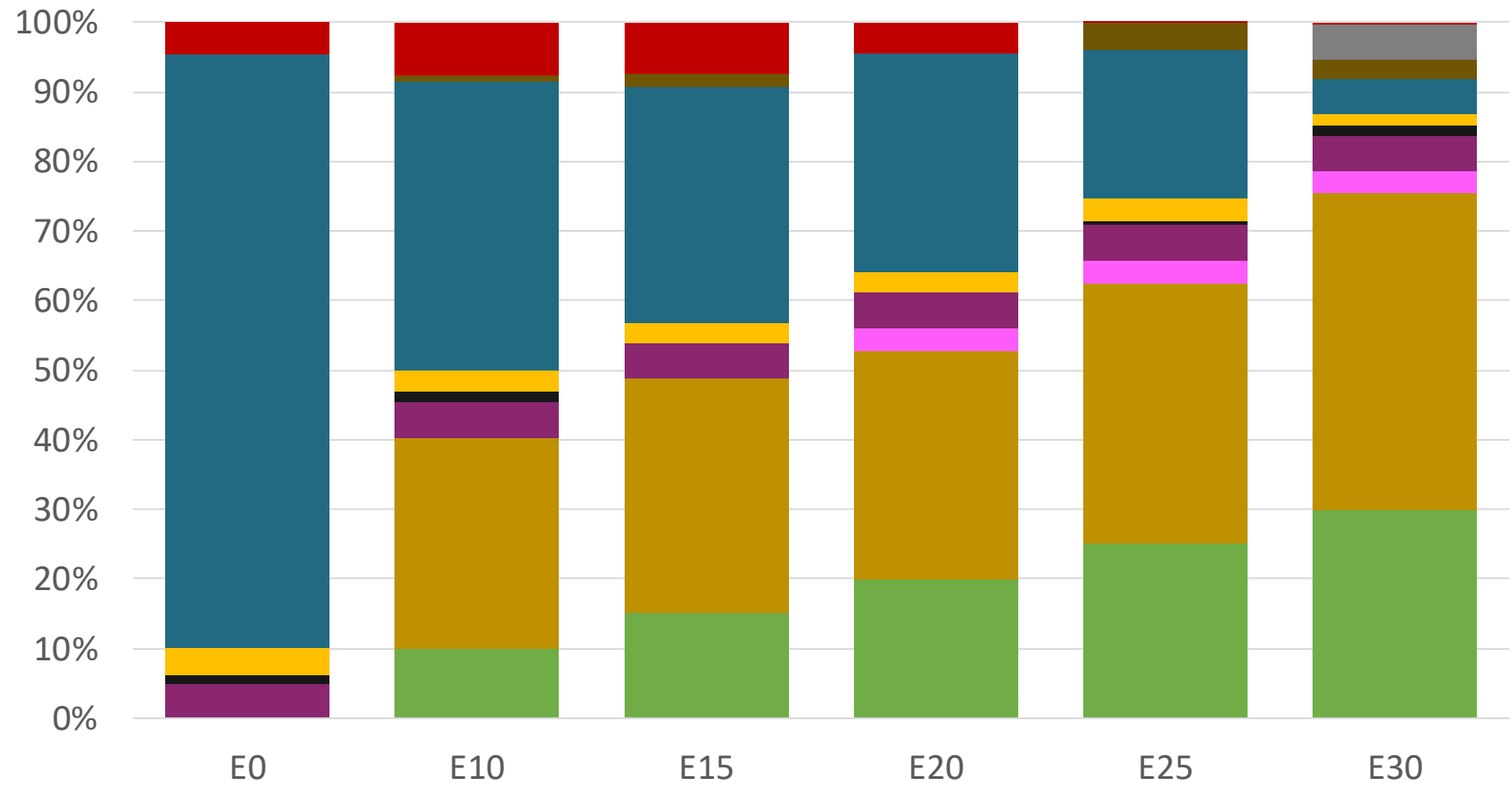
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.156	\$ 1.885	\$ 1.734	\$ 1.592	\$ 1.447	\$ 1.477

Honduras – Premium – Constant Octane



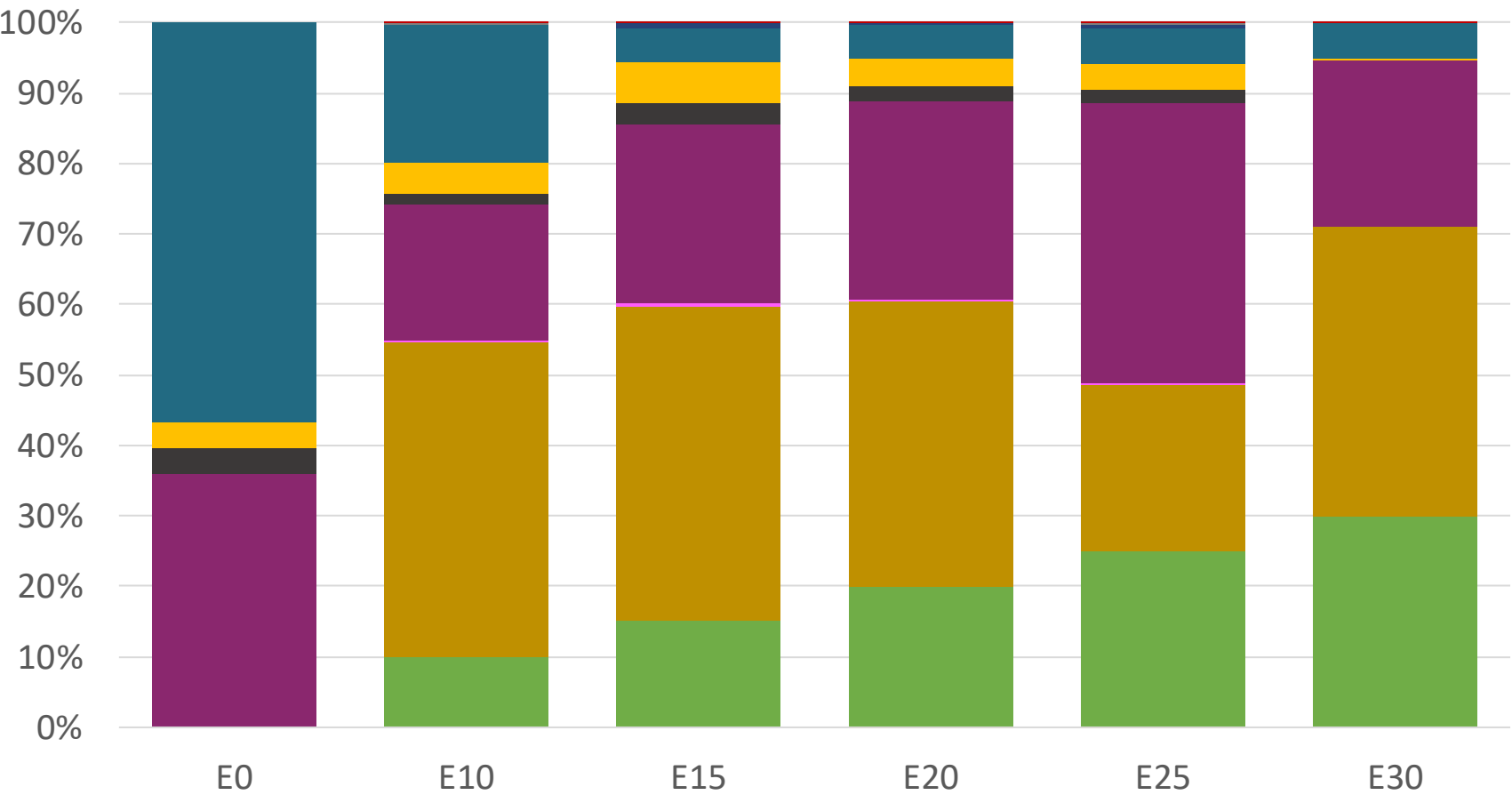
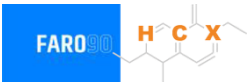
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.536	\$ 2.233	\$ 2.099	\$ 1.957	\$ 1.807	\$ 1.659

Honduras – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156

Honduras – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

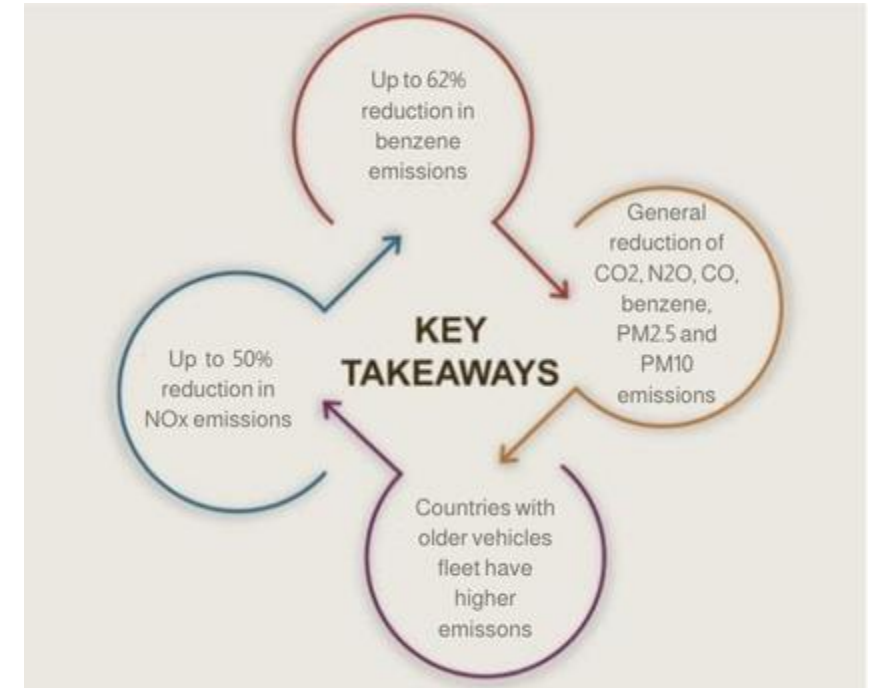
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

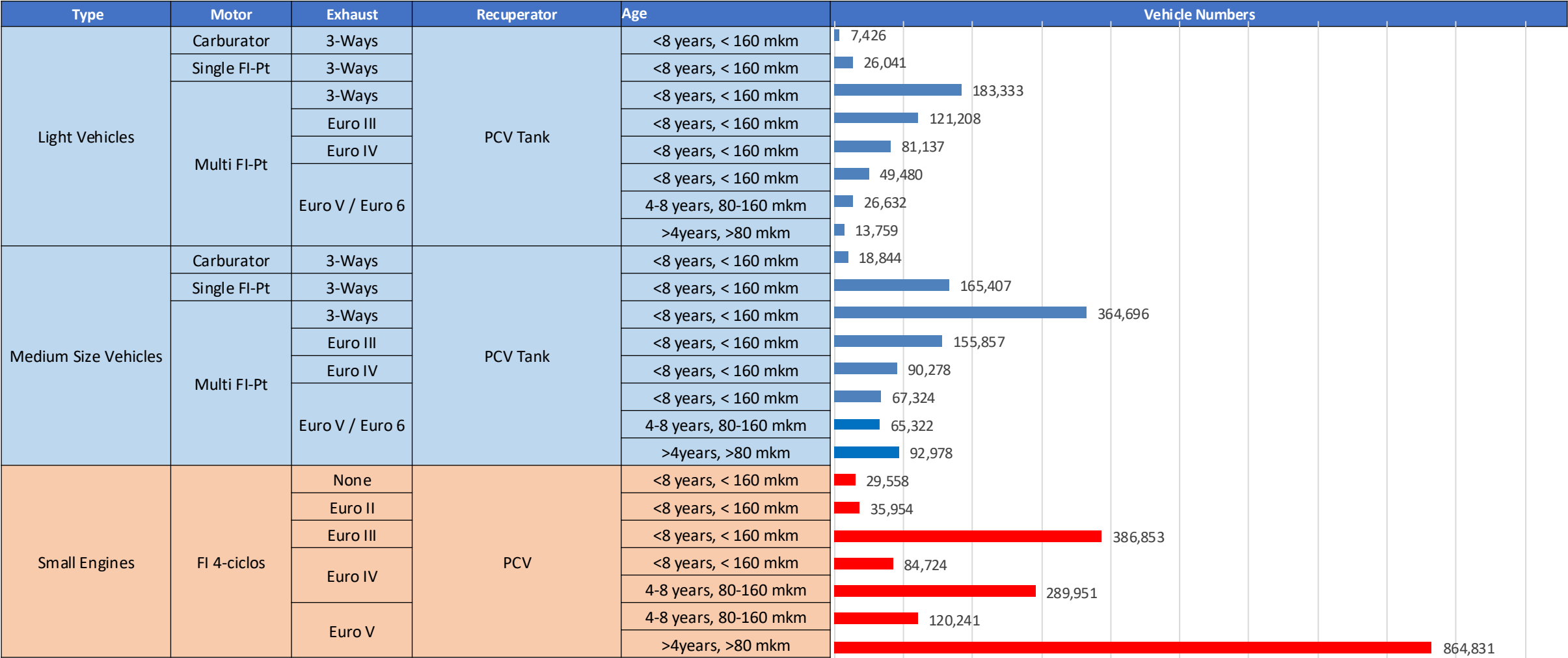
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

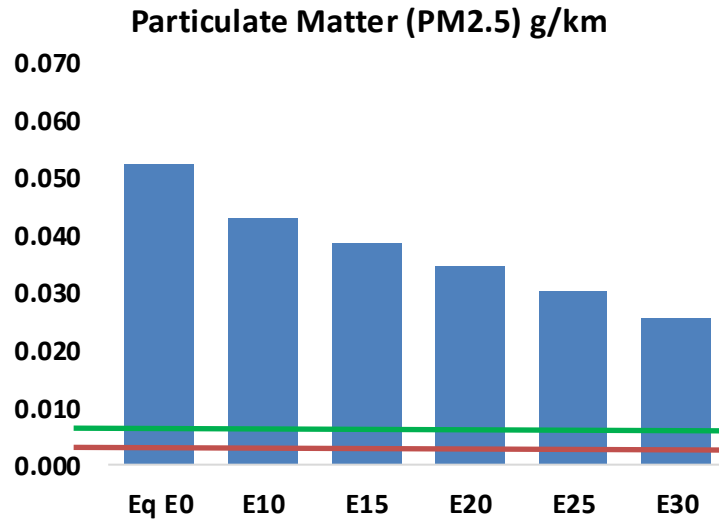
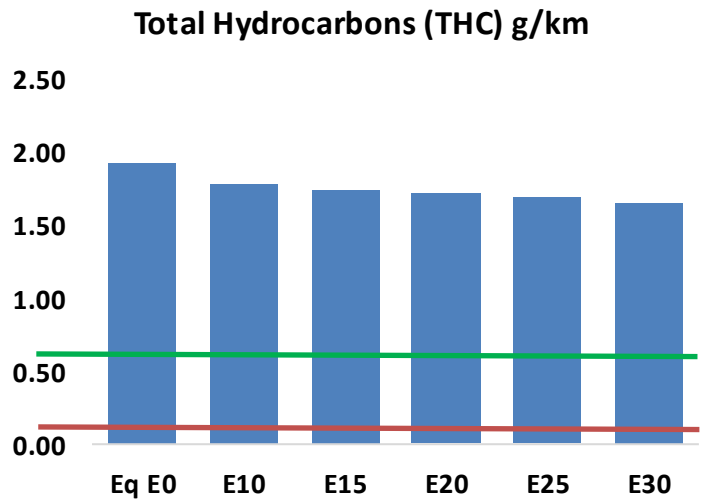
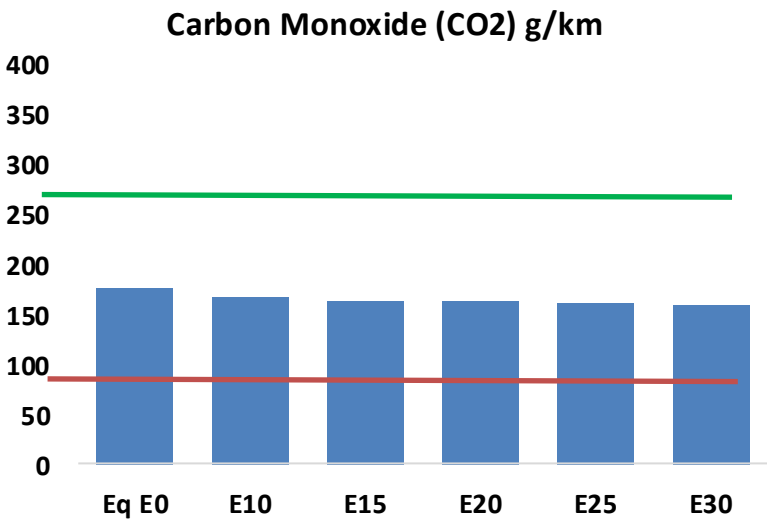
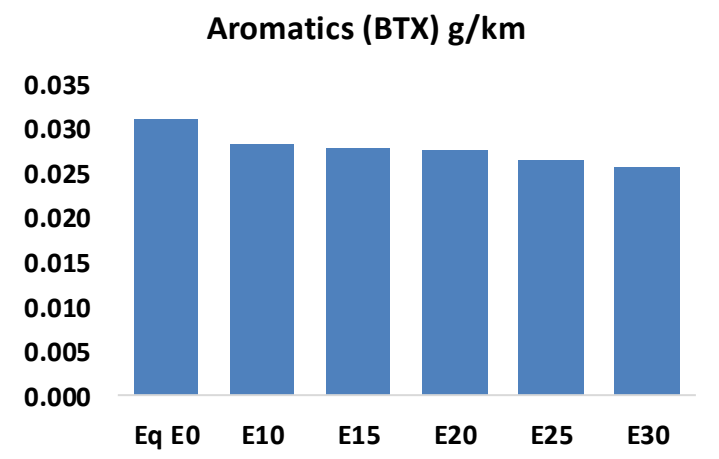
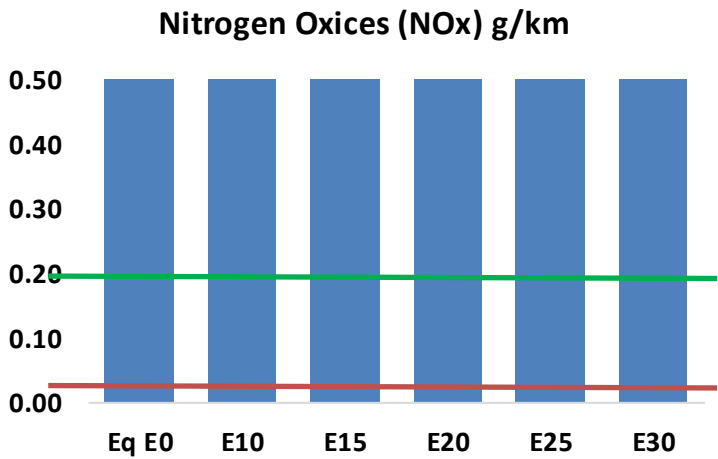
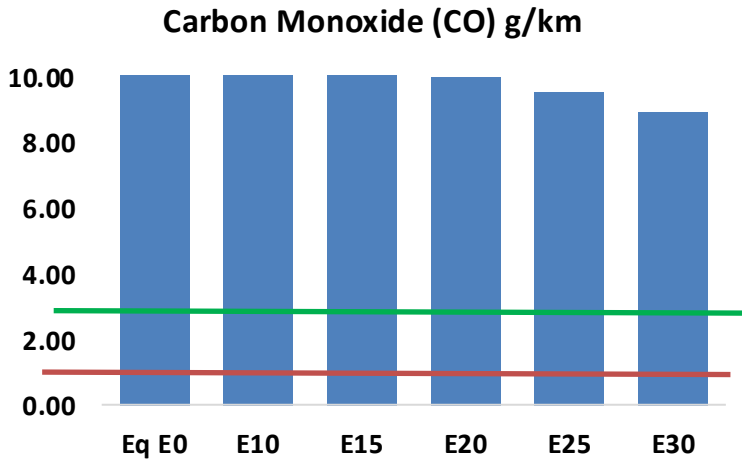
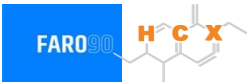
Gasoline Vehicular Fleet in Honduras



Vehicular Fleet: **3,341,832**
Average Age: **13 years**
Motorcycles: **50%**

Honduras – Vehicular Emissions

United States
Euro 6



Honduras – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	769,641	720,753	671,866	633,811	596,439	567,980	532,623	-13%	-23%
CO2	10,458,718	10,189,779	9,935,782	9,736,021	9,637,171	9,612,562	9,435,375	-5%	-8%
VOC	114,270	110,163	106,056	103,603	101,554	100,106	97,726	-7%	-11%
NOx	48,100	44,412	41,946	40,724	38,717	37,162	35,319	-13%	-20%
SOx	123	113	104	96	89	81	72	-15%	-28%
NH3	4,189	4,148	4,106	4,126	4,172	4,205	4,232	-2%	0%
N2O	192	191	190	191	195	197	199	-1%	2%
CH4	25,506	25,506	25,506	25,889	26,450	26,934	27,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	434	406	378	356	334	318	295	-13%	-23%
Acetaldehyde	1,087	1,248	1,831	2,788	3,799	4,341	5,129	68%	249%
Formaldehyde	4,114	4,000	4,663	5,475	5,736	6,346	6,908	13%	39%
Benzene	1,849	1,776	1,683	1,657	1,643	1,571	1,520	-9%	-11%
PM2.5	619	563	507	457	409	357	302	-18%	-34%
PM10	2,490	2,264	2,037	1,838	1,643	1,434	1,216	-18%	-34%

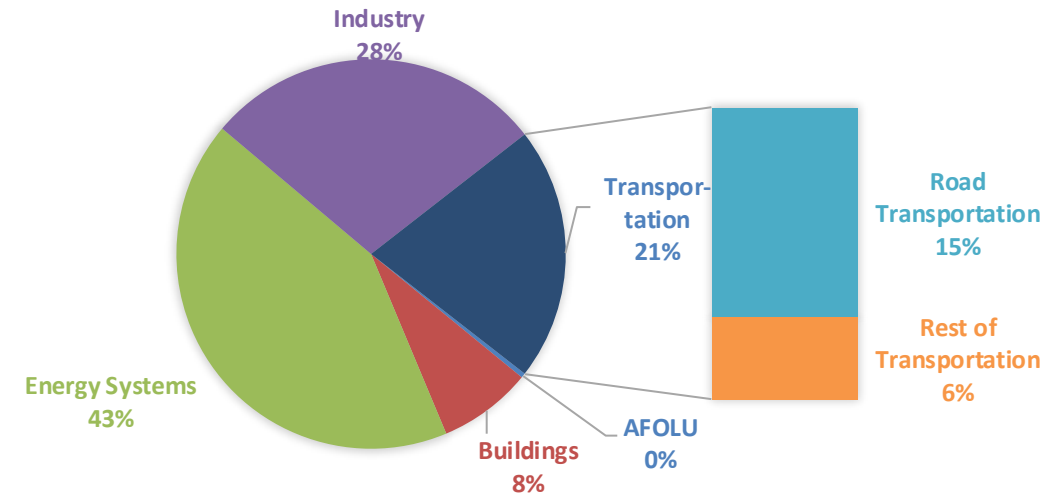
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

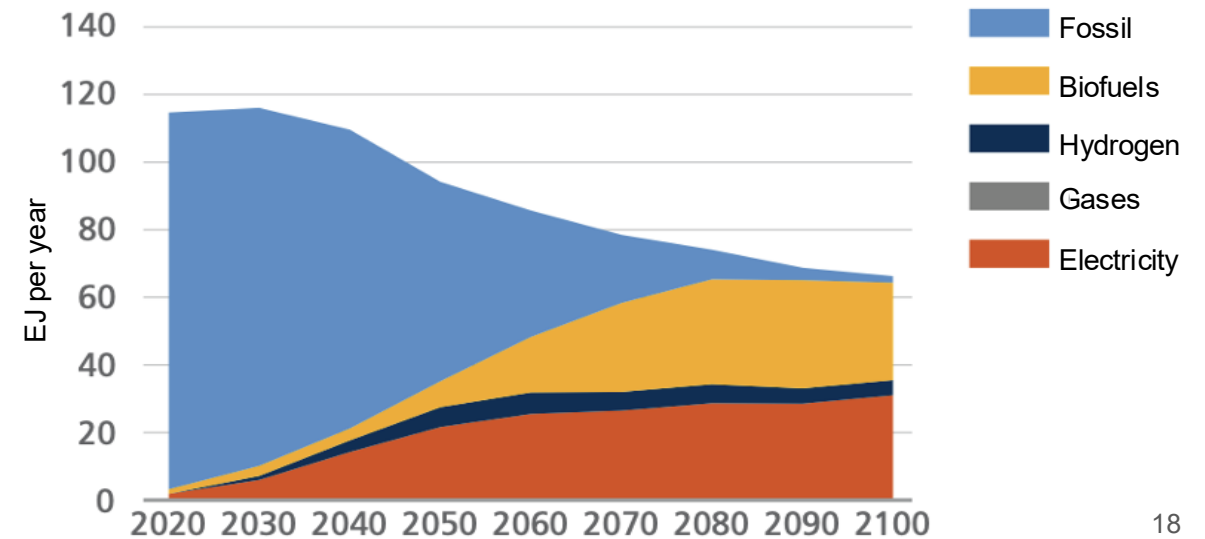
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

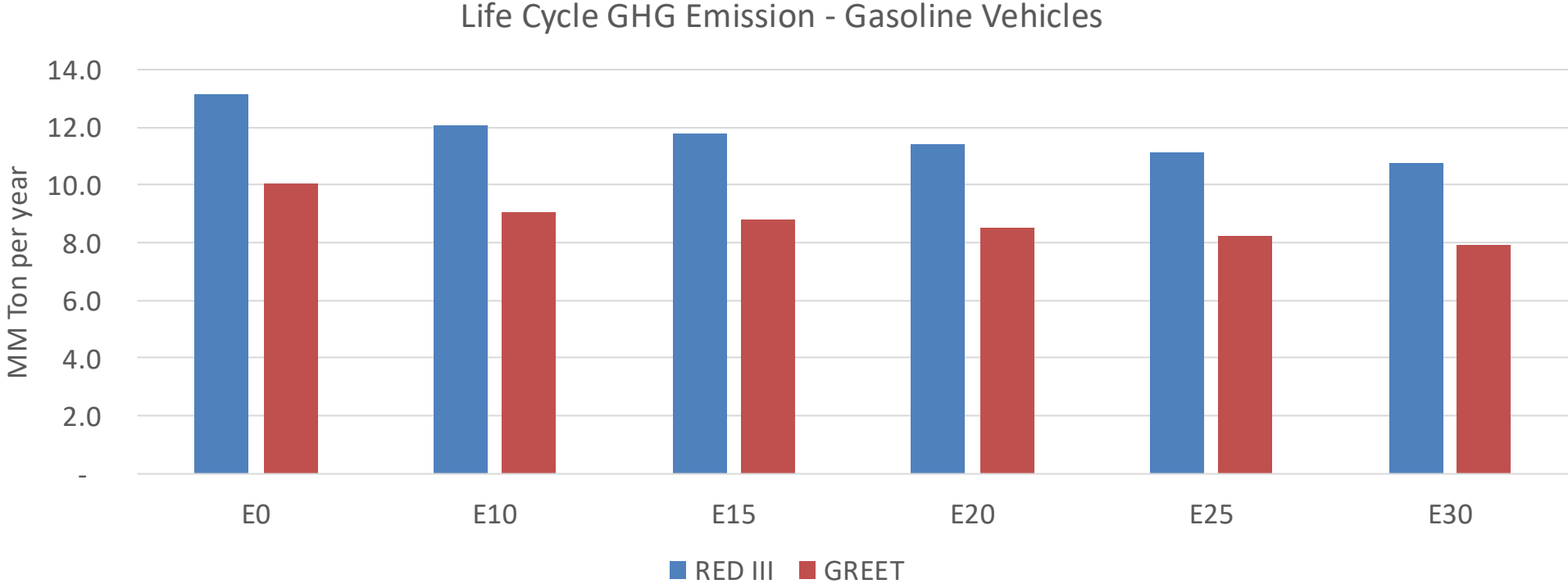
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Honduras – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	30.7
2035 Target (MMT/year)	15.3
Delta	15.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.8%	12.4%	15.6%	18.0%	21.3%
RED II/III	0.0%	8.2%	10.6%	13.3%	15.5%	18.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.4%	8.2%	10.2%	11.8%	14.0%
RED II/III	0.0%	7.1%	9.1%	11.4%	13.3%	15.7%